

CHAPTER 6 DRILLS

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D6.1

Given data:

$$\epsilon_r = 3.8$$

$$D = 8 \frac{nC}{m^2}$$

$$(a) E = \frac{D}{\epsilon_o \epsilon_r} = \underline{237.7 \frac{V}{m}}$$

(b) We first find χ_e :

$$\chi_e = \epsilon_r - 1 = 2.8$$

Now,

$$P = \chi_e \epsilon_o E = \underline{5.9 \frac{nC}{m^2}}$$

(c) We know that,

$$\frac{No. of Dipoles}{Volume} = \frac{P}{Average Dipole Moment} = \frac{5.9 \times 10^{-9}}{10^{-29}} = \underline{5.9 \times 10^{20} m^{-3}}$$

D6.2

Given Data:

$$\epsilon_{r1} (z < 0) = 3.2$$

$$\epsilon_{r2} (z > 0) = 2$$

$$D_1 = -30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z \frac{nC}{m^2}$$

$$(a) D_{N1} = D_1 \cdot z = \underline{70 \frac{nC}{m^2}}$$

(b) The tangential vector components of $\mathbf{D}_1$ are perpendicular to z -axis, so:

$$\mathbf{D}_{t1} = \underline{-30\mathbf{a}_x + 50\mathbf{a}_y \frac{nC}{m^2}} \quad (\because x \text{ and } y \text{ axes are } \perp)$$

(c) Magnitude of $\mathbf{D}_{t1}$ is:

$$D_{t1} = \sqrt{30^2 + 50^2} = \underline{58.3 \frac{nC}{m^2}}$$

(d) Magnitude of $\mathbf{D}_1$ is:

$$D_1 = \sqrt{30^2 + 50^2 + 70^2} = \underline{91.1 \frac{nC}{m^2}}$$

$$(e) \theta_1 = \cos^{-1} \left(\frac{D_{N1}}{D_1} \right) = \underline{39.8^\circ}$$

$$(f) \mathbf{P}_1 = \chi_e \epsilon_o \mathbf{E}_1 = (\epsilon_{r1} - 1) \frac{\mathbf{D}_1}{\epsilon_{r1}} = \underline{-20.6\mathbf{a}_x + 34.4\mathbf{a}_y + 48.1\mathbf{a}_z \frac{nC}{m^2}}$$

D6.3

Given Data:

From previous drill problem!

(a) $D_{N2} = D_{N1} = 70 \frac{nC}{m^2}$, so, $\mathbf{D}_{N2} = \underline{70\mathbf{a}_z \frac{nC}{m^2}}$

(b) As, $\frac{D_{t1}}{D_{t2}} = \frac{\epsilon_{r1}}{\epsilon_{r2}} \Rightarrow \mathbf{D}_{t2} = \frac{\epsilon_{r2} \times \mathbf{D}_{t1}}{\epsilon_{r1}} = \underline{-18.75\mathbf{a}_x + 31.25\mathbf{a}_y \frac{nC}{m^2}}$

(c) $\mathbf{D}_2 = \mathbf{D}_{N2} + \mathbf{D}_{t2} = -18.75\mathbf{a}_x + 31.25\mathbf{a}_y + 70\mathbf{a}_z \frac{nC}{m^2}$:

(d) $\mathbf{P}_2 = \chi_e \epsilon_o \mathbf{E}_2 = (\epsilon_{r2} - 1) \frac{\mathbf{D}_2}{\epsilon_{r2}} = \underline{-9.38\mathbf{a}_x + 15.63\mathbf{a}_y + 35\mathbf{a}_z \frac{nC}{m^2}}$

(e) $\theta_2 = \cos^{-1} \left(\frac{D_{N2}}{D_2} \right) = \underline{27.5^\circ}$

D6.4

(a) We first find C:

$$C = \frac{2W_E}{V_o^2} = 13.9 \mu F$$

Now,

$$\epsilon_r = \frac{Cd}{S\epsilon_o} = \underline{1.04}$$

(b) Here, $W_E = 100 \times d \times S = 100 \times 45 \mu \times 0.12 = 5.4 \times 10^{-4} J$

So, $C = \frac{2W_E}{V_o^2} = 27 \mu F$.

$$\epsilon_r = \frac{Cd}{S\epsilon_o} = \underline{1.14}$$

(c) As, $E = \frac{V}{d} \Rightarrow \epsilon_r = \frac{Cd}{E\epsilon_o} = \underline{1.13}$

D6.5

$$(a) C = \frac{2\pi\epsilon L}{\ln(b/a)} = \frac{2\pi\epsilon_0(2.26)(1 \times 30 \times 10^{-2})}{\ln(0.680/0.1045)} = \underline{20.05 pF}$$

$$(b) C = \frac{4\pi\epsilon}{\frac{1}{a} + \frac{1}{b}} = \frac{4\pi\epsilon_0(2.26)}{\frac{1}{2.5m} + \frac{1}{4.5m}} = \underline{1.41 pF}$$

(c) For each of three plates, $S = 1 cm \times 4 cm = 4 \times 10^{-4} m^2$ and $d = 0.1 mm$, so

$$C = \frac{1}{\frac{d}{\epsilon_1 S} + \frac{d}{\epsilon_2 S} + \frac{d}{\epsilon_3 S}} = \underline{28.71 pF}$$

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D6.6

Given Data:

$$b = 1 cm$$

$$V_o = 20 V$$

$$h = 5 cm$$

$$\epsilon_r = 4.5$$

$$(a) C = \frac{2\pi\epsilon}{\cosh^{-1}(h/b)} = \underline{109.205 pF}$$

$$(b) \rho_{s,max} = \frac{\rho_L}{2\pi} \left[\frac{h-b+a}{(h-b+a)^2} - \frac{h-b-a}{(h-b-a)^2} \right], \text{ so we first find } \rho_L \text{ and } a:$$

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$$\rho_L = \frac{4\pi\epsilon V}{\ln(K_1)}, \text{ where } \sqrt{K_1} = \frac{h + \sqrt{h^2 - b^2}}{b} = 9.89, \text{ so } K_1 = 97.98$$

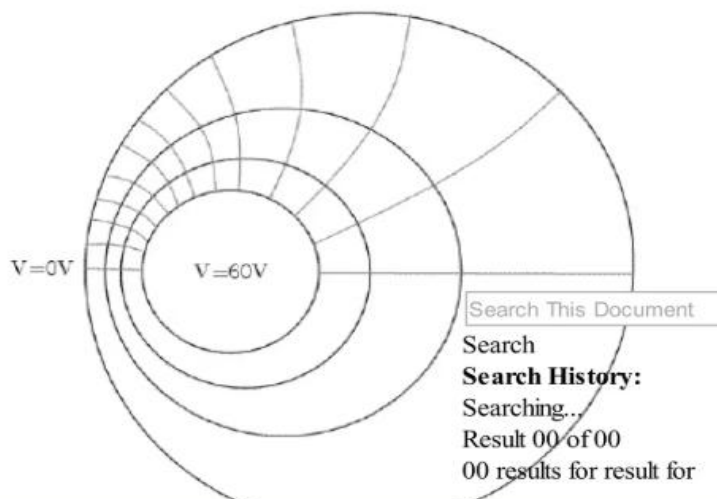
$$\text{Now, } \rho_L = 2.18 \pi \epsilon, \text{ and}$$

$$a = \sqrt{h^2 - b^2} = 0.049$$

Now,

$$\rho_{s,max} = 3.46 \times 10^{-10} \left[\frac{0.089}{(0.089)^2} - \frac{-9 \times 10^{-3}}{(-9 \times 10^{-3})^2} \right] = \underline{42.33 \frac{nC}{m^2}}$$

D6.7



$$(a) C = \epsilon_o \frac{N_O}{N_V} = \epsilon_o \frac{2 \times 12}{3} = \underline{\underline{70.8 \frac{pF}{m}}} \boxed{1}$$

$$(b) E = \frac{\Delta V}{\Delta L_N} = \frac{20V}{\frac{1mm}{3}} = 60 \frac{kV}{m} \boxed{2}$$

$$(c) \rho_S = \epsilon E = \epsilon_o (60k) = \underline{\underline{531 \frac{nC}{m^2}}}$$